



Project (20 marks)

1. Introduction

In this project, you will work in teams of 2 to 3 students. You will be required to turn in both your code, and a detailed report for the work done step by step and results of your project. In addition, you are required to pass the final discussion session of your work as a team before the final exam. You can choose one case for A and B for your documentation.

2. Project Outline

Design and implement a 3-bit Arithmetic and Logic Unit (ALU) that has inputs A & B to do the following:

Operation	Output (F)
ADD	$F=A + B$
SUB	$F=B - A$
2'S CMP	$F=A' + 1$
OR	$F=A \mid B$
AND	$F=A \& B$
XOR	$F=A \wedge B$
Not	$F=B'$
Shift Left Logical (for Unsigned Numbers)	$F=A \ll 2$
Shift Right Logical (for Unsigned Numbers)	$F=A \gg 3$

Bonus (3 marks)

The ALU should also have the following output bits:

- Output bit V, where $V = 1$ in case of an overflow. Also, Output bit Z, where $Z=1$ if $F=0$.
- Shift Left and Shift Right (for Arithmetic Signed and Unsigned Numbers).
- Testbench for the ALU design using SystemVerilog.

3. Deliverables

1. A clear and detailed report documenting:

- The ALU design block using LogiSim.
- Ensure that the ALU Implemented design clears all inputs, outputs, and internal wires for selecting which block of the previously mentioned operations (Do a figure for the ALU that explains each block integrated inside in detail).
- The ModelSim simulation outputs by forcing mechanism (with clear description).

2. HDL code files using SystemVerilog.